Visual Scene Understanding through Neuro-Symbolic Reasoning

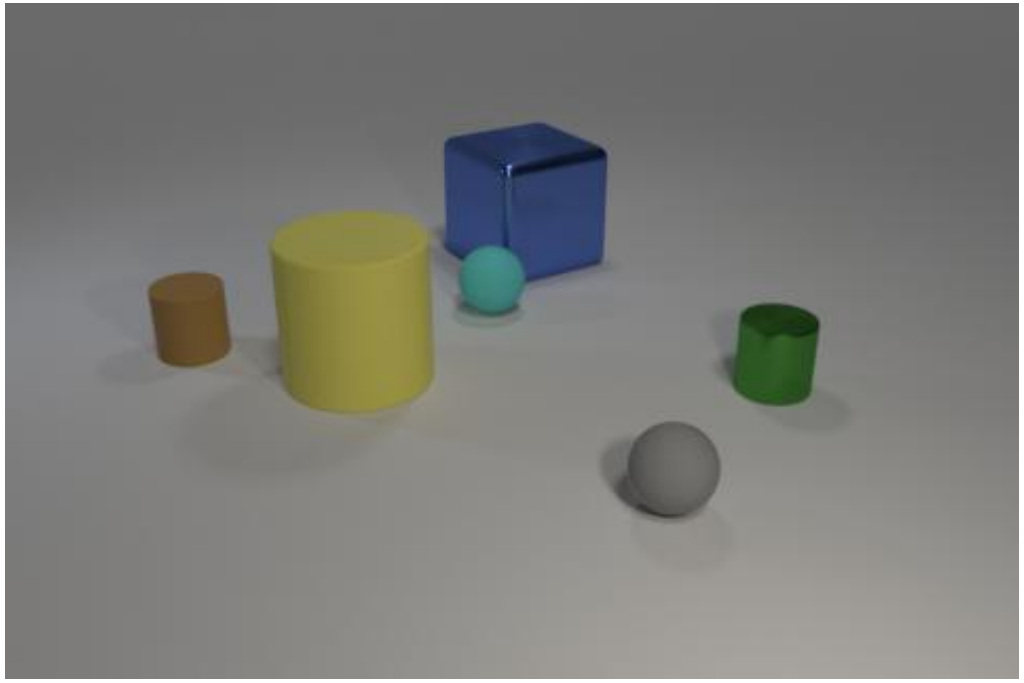
Bachelor's Thesis

Bogdan Đukić, RA98/2021

Supervisor: Dr. Milan Segedinac

University of Novi Sad, Faculty of Technical Sciences

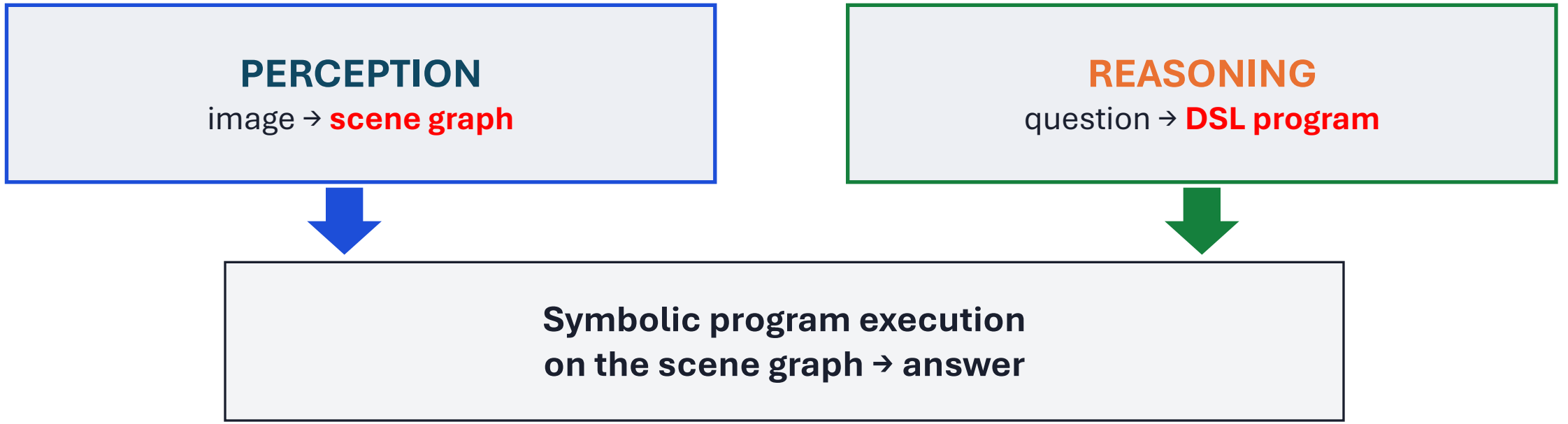
Problem: Visual Question Answering (VQA)



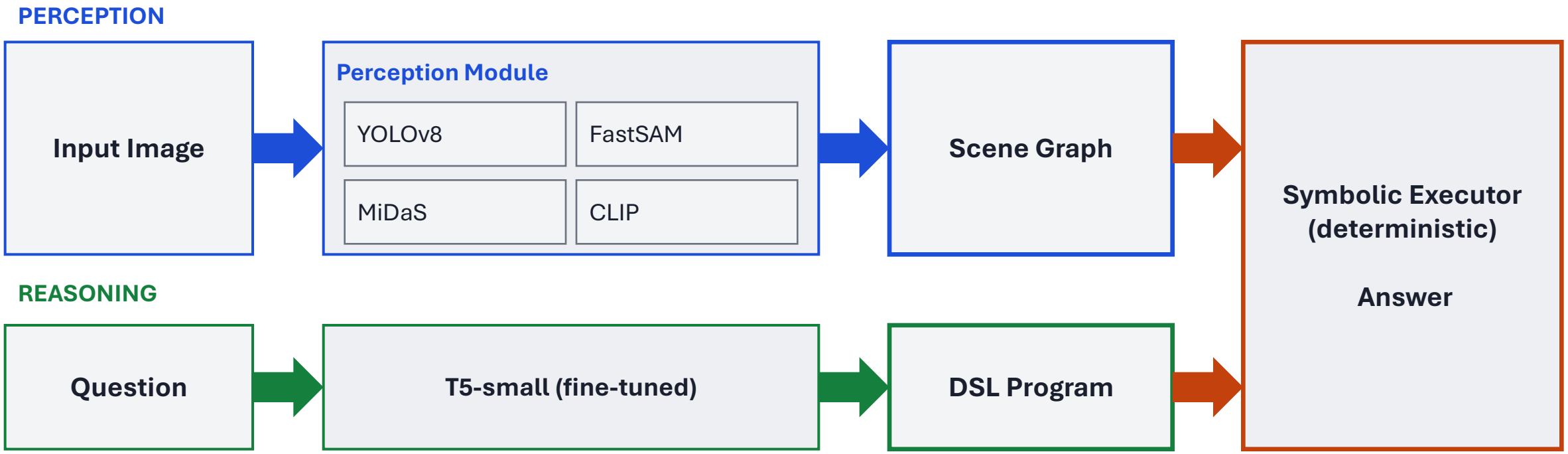
Question: "How many cylinders are there in the image?" →
Answer: 3

- Input: image + natural-language question → Output: answer (number / yes-no / attribute)
- End-to-end neural models: high accuracy, but a **black box**
- **The problem: when a model makes a mistake—where did the error occur?**

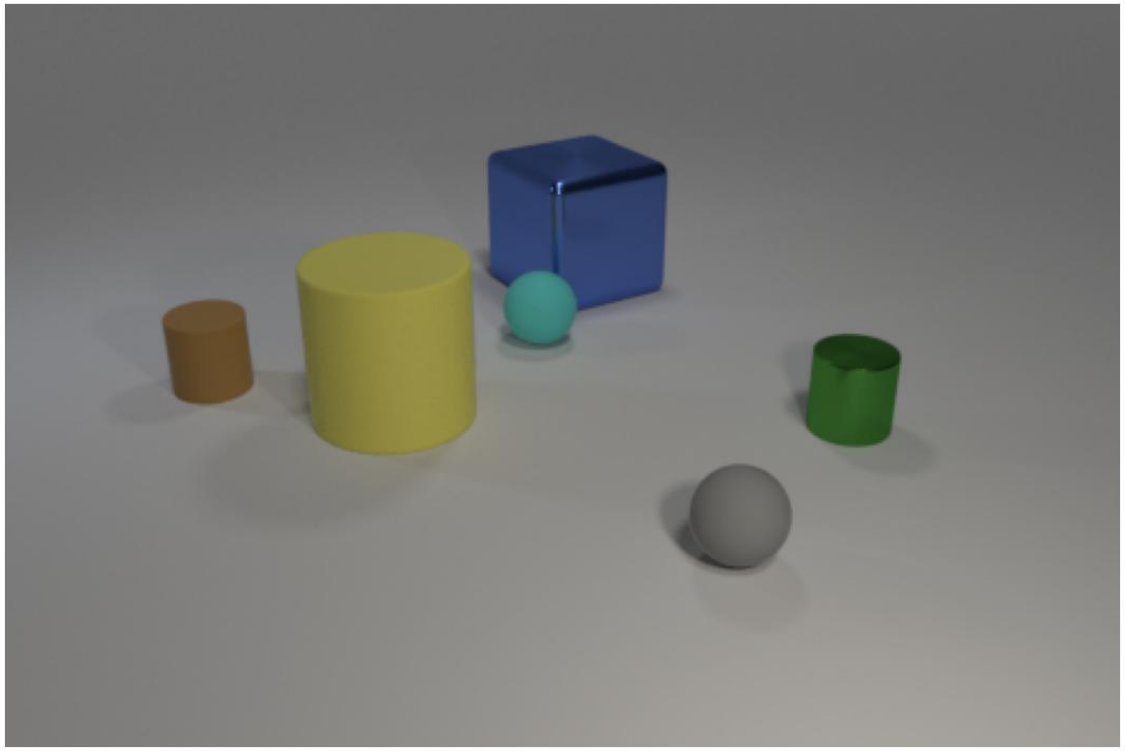
Solution: A Neuro-Symbolic Approach



System Architecture



System Input

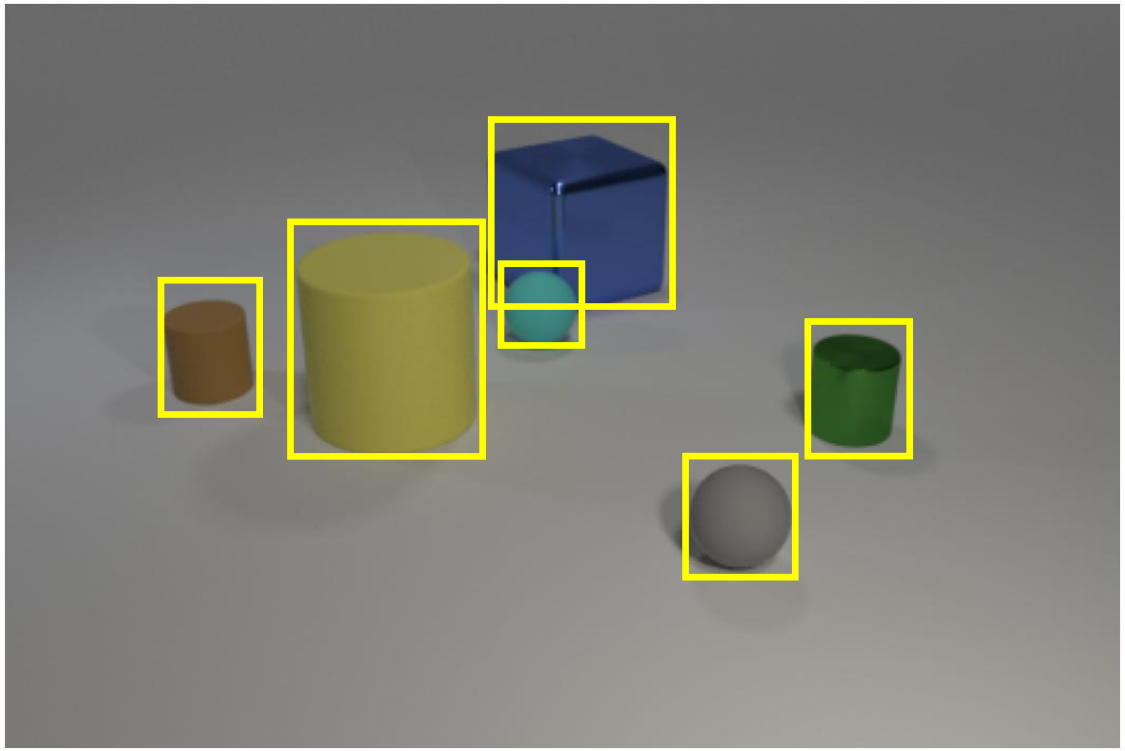


Question: "What color is the cube behind the cyan sphere?"

Input Image

The system receives an image of a scene and a natural-language question. The expected answer is a number, yes/no, or an attribute value.

Object Detection



Question: "What color is the cube behind the cyan sphere?"

YOLOv8m × FastSAM

Hybrid Detection

YOLOv8 detects the bounding boxes of all six objects; FastSAM in "segment everything" mode supplements detections missed by YOLO. The results are merged using intersection over union (IoU).

Depth Estimation



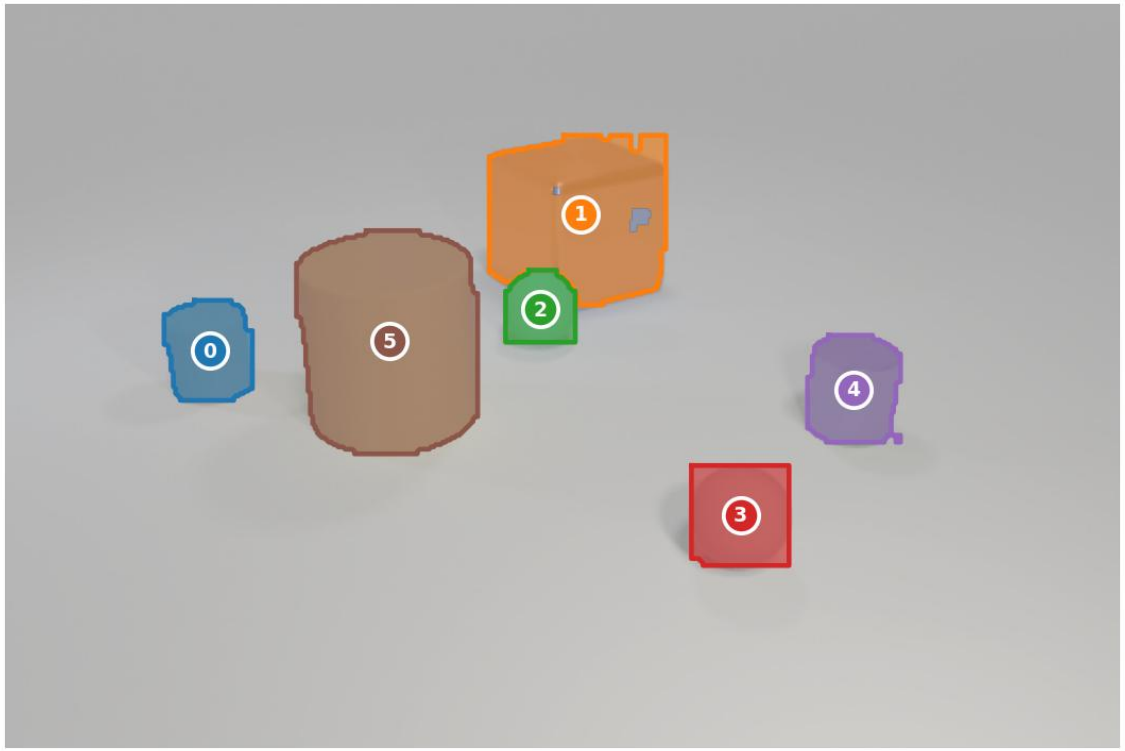
Question: "What color is the cube behind the cyan sphere?"

MiDaS

Relative Depth Map

MiDaS estimates relative depth from a single image: brighter regions are closer. Depth helps determine front/behind relationships and resolve overlapping masks in the next step.

Masks and Z-Buffer



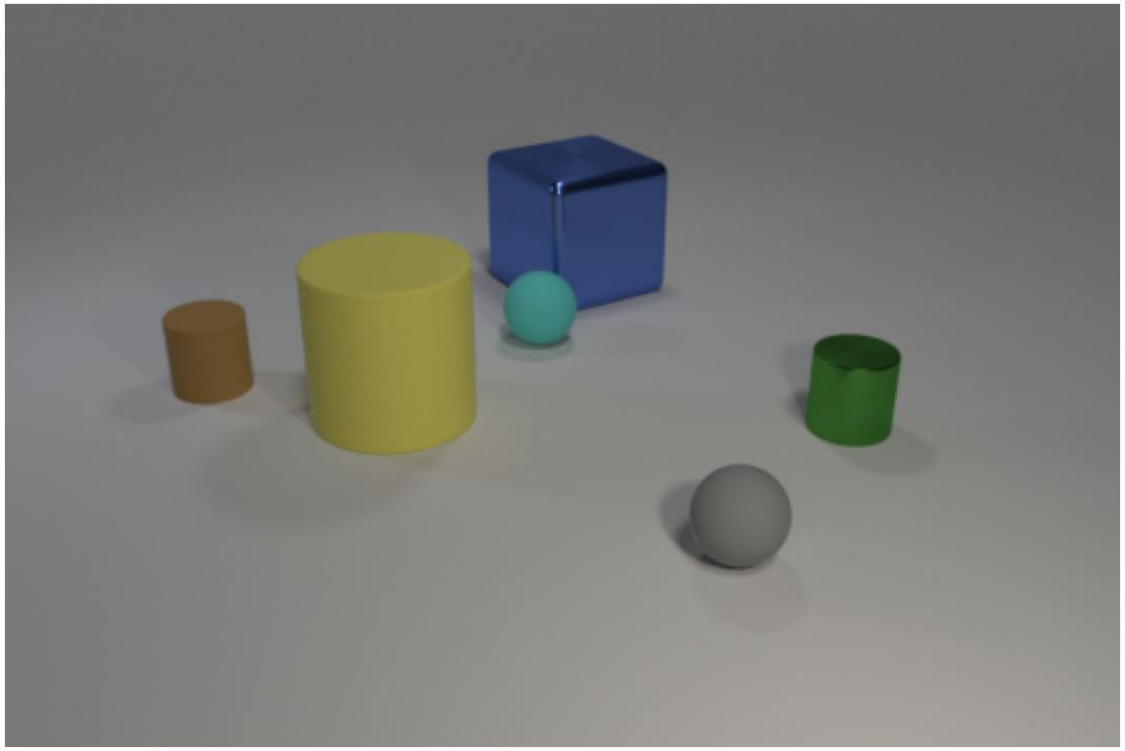
Question: "What color is the cube behind the cyan sphere?"

FastSAM × Z-Buffer

Non-overlapping Segmentation

A precise mask is extracted for each bounding box. The Z-buffer assigns each pixel to exactly one object.

Attributes → Scene Graph



Question: "What color is the cube behind the cyan sphere?"

CLIP (zero-shot)

Attributes → Scene Graph

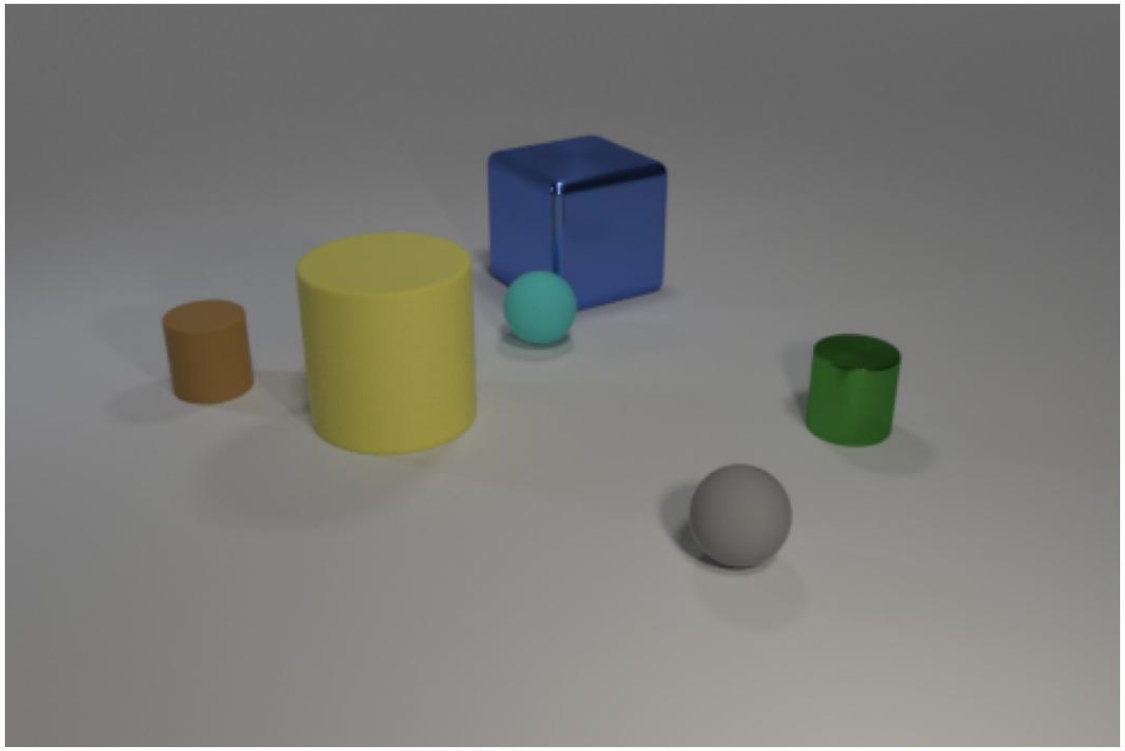
CLIP classifies each object's color, shape, and material.

```
objects:
  0  brown   cylinder  rubber  S
  1  yellow  cylinder  rubber  L
  2  blue    cube      metal   L
  3  cyan    sphere    rubber  S
  4  green   cylinder  metal   S
  5  gray    sphere    rubber  S

relationships:
  behind[3] = [2] ← behind the cyan sphere
  left[2]   = [0, 1]
  right[3]  = [4, 5]
 …
```



Question → DSL Program



Question: "What color is the cube behind the cyan sphere?"

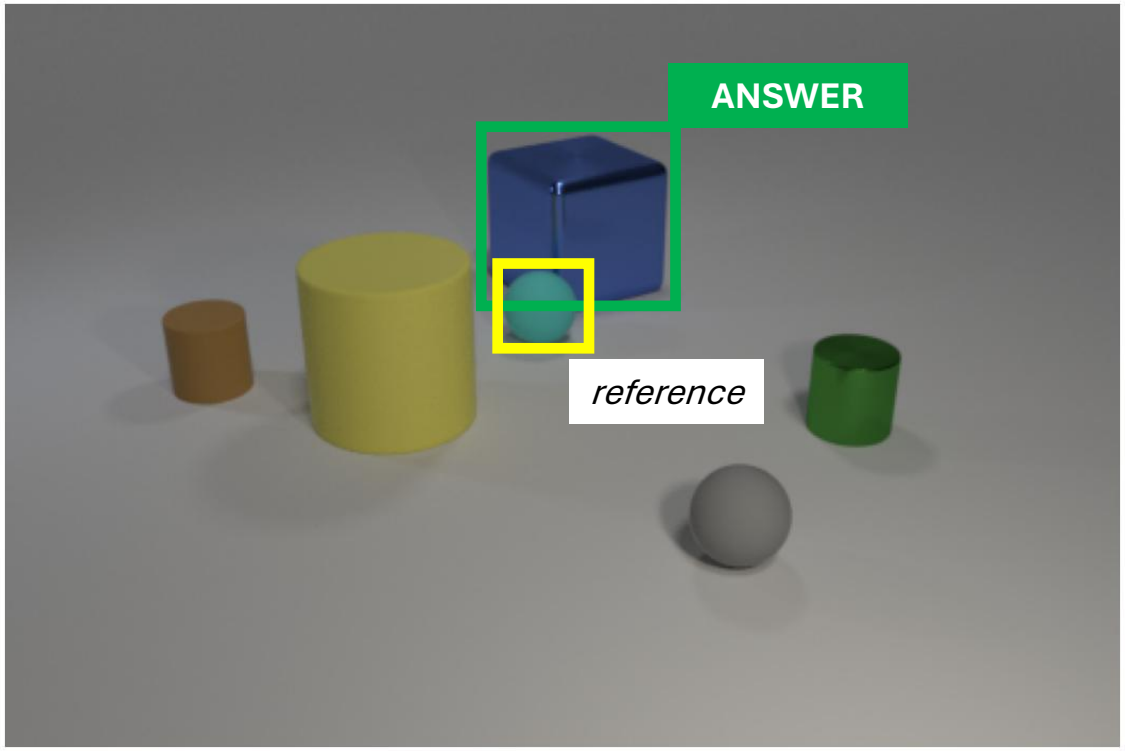
T5-small (fine-tuned)

Question → DSL Program

```
1  scene
2  filter_color[cyan]
3  filter_shape[sphere]
4  unique
5  relate[behind]
6  filter_shape[cube]
7  unique
8  query_color
```



Symbolic Execution



Question: "What color is the cube behind the cyan sphere?"

Executor - deterministic

The program is executed step by step

```
1  scene
2  filter_color[cyan]
3  filter_shape[sphere]
4  unique
5  relate[behind]
6  filter_shape[cube]
7  unique
8  query_color
```

Answer: “blue”

Experimental Setup

CLEVR
Synthetic scenes and template-generated questions - used for training and baseline evaluation.

CLEVR-CoGenT
Previously unseen attribute combinations not found in training - tests compositional generalization.

CLEVR-Humans
Natural-language questions written by humans, including synonyms.

Two Evaluation Pipelines

GT Pipeline (Reference)
Ground-Truth Scene Graph → Program → Answer
Isolates reasoning performance

E2E Pipeline (Full System)
Image → Generated Scene Graph → Program → Answer

Question-to-DSL Translation (Reference Pipeline)

Dataset	Accuracy	Valid Programs	Execution Failures
CLEVR (reference)	**99.9%**	**100.0%**	**0.0%**
CLEVR-CoGenT valB	**99.7%**	**100.0%**	**0.1%**
CLEVR-Humans (before RL)	**41.4%**	**90.5%**	**25.5%**
CLEVR-Humans (after RL)	**57.1%**	**92.6%**	**0.6%**

REINFORCE (without any program annotations from CLEVR-Humans)

Accuracy: **41.4%** → **57.1%** (+15.7 pp)

Execution failures: **25.5%** → **0.6%** (−24.9 pp)

Direct Perception Evaluation: Error Localization

Detection	Value
Recall	**97.5%**
Precision	**88.7%**
F1 Score	**92.9%**
FN / image	**0.16**
FP / image	**0.80**

CLIP Attributes	Accuracy
Shape	**95.5%**
Material	**94.4%**
Size	**92.8%**
Color	**87.8%**

Spatial Relations	Accuracy
front	**68.6%**
right	**66.7%**
left	**66.5%**
behind	**44.3%**
Overall	**61.5%**

green	blue	red	yellow	gray	brown
97%	**91%**	**90%**	**87%**	**86%**	**71%**

What Is the Cost of Perception?

Dataset	GT Pipeline	E2E Pipeline	Gap (pp)
CLEVR val	**99.9%**	**79.0%**	**−20.9**
CoGenT valB	**99.7%**	**79.0%**	**−20.7**
CLEVR-Humans (before RL)	**41.4%**	**32.7%**	**−8.7**
CLEVR-Humans (after RL)	**57.1%**	**45.6%**	**−11.5**

Sanity check - a typical program filters by color, shape, and material:

P(all 3 filters correct) ≈ 0.878 × 0.955 × 0.944 ≈ **79.2%**

Measured E2E accuracy: **79.0%** ✓

Comparison with IEP and NS-VQA

Dataset	IEP	NS-VQA	Proposed - E2E	Proposed - GT
CLEVR val	96.9%	99.8%	79.0%	**99.9%**
CoGenT B	73.7%	**63.9%**	**79.0%**	**99.7%**
CLEVR-Humans	66.6%	67.0%	45.6%	**57.1%**

GT pipeline matches leading neuro-symbolic systems (99.9% vs. 99.8% for NS-VQA).

On CoGenT, our E2E system outperforms both methods - separate attributes in the DSL ≠ learned correlations (NS-VQA: 48 combined classes → 63.9%).

Conclusions and Future Work

Three Key Findings

- Neuro-symbolic reasoning: **~99.9%** accuracy given a correct scene graph
- Key advantage: **errors can be localized by stage** (color and spatial relations)
- RL adapts to natural language **without program annotations** (+15.7 pp)

Future Work

- Dedicated color classifier
- Improved depth estimation for spatial relationships
- Extend the DSL (superlatives, “between,” most frequent attribute)

Thank You!

Questions?